

A Redox-Active Supramolecular Fe_4L_6 Cage Based on Organic Vertices with Acid–Base-Dependent Charge Tunability for Dehydrogenation Catalysis

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Cite This: *J. Am. Chem. Soc.* 2022, 144, 8778–8788



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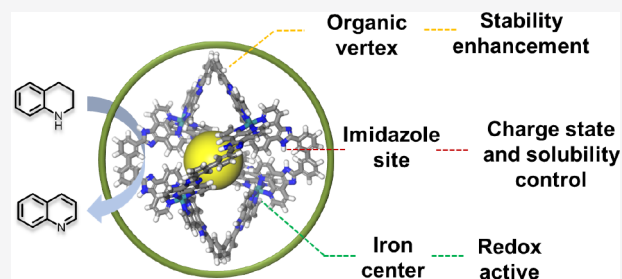


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ABSTRACT: Supramolecular cage chemistry is of lasting interest because, as artificial blueprints of natural enzymes, the self-assembled cage structures not only provide substrate-hosting biomimetic environments but also can integrate active sites in the confined nanospaces for function synergism. Herein, we demonstrate a vertex-directed organic-clip chelation assembly strategy to construct a metal–organic cage $\text{Fe}_4\text{L}_6^{8+}$ (MOC-63) incorporating 12 imidazole proton donor–acceptor motifs and four redox-active Fe centers in an octahedral coordination nanospace. Different from regular supramolecular cages assembled with coordination metal vertices, MOC-63 comprises six ditopic organic-clip ligands as vertices and four tris-chelating $\text{Fe}(\text{N}\text{ON})_3$ moieties as faces, thus improving its acid, base, and redox robustness by virtue of cage-stabilized dynamics in solution. Improved dehydrogenation catalysis of 1,2,3,4-tetrahydroquinoline derivatives is accomplished by MOC-63 owing to a supramolecular cage effect that synergizes multiple Fe centers and radical species to expedite intermediate conversion of the multistep reactions in a cage-confined nanospace. The acid–base buffering imidazole motifs play a vital role in modulating the total charge state to resist pH variation and tune the solubility among varied solvents, thereby enhancing reaction acceleration in acidic conditions and rendering a facile recycling catalytic process.



INTRODUCTION

Supramolecular catalysis of artificial synthetic hosts is bioinspired for properties and functions intrinsic to enzymes but going far beyond mimics,^{1–4} which has promoted the development of the construction strategies of metal–organic cages (MOCs), representing one type of synthetic supramolecular cage based on noncovalent self-assembly, toward maturity in the past decades.^{5–10} For example, the well-known Fujita's molecular paneling,^{11–15} Stang's molecular library,^{16–18} Nitschke's subcomponent self-assembly,¹⁹ and many other strategies have derived a plenty of cage structures.^{10,20–29} In most cases, the MOCs are formed with the metal sites acting as polyhedral vertices, where the metal–ligand coordination bonds might not be robust enough to bear tough reaction conditions, especially when monodentate M–N or dynamic C≡N bonds are involved in cage assembly. As the research interest in this field moves rapidly forward to the functionalities such as catalysis,^{30,31} separation,^{32–34} photoluminescence,^{35–38} and bioapplications,^{39–41} there is an immediate need for the design and synthesis of MOCs that can resist harsh acidic, basic, and redox conditions.

To utilize MOCs as enzyme-mimicking supramolecular flasks,⁴² most works focus on their catalytic performance enforced by the confinement effect of the host cavities and the charge state of the cages,^{9,43–46} while little attention is paid to

the potential of metal centers to act as functional sites that actually present the pivotal active component in natural enzyme catalysis. A few examples with catalytic activity arisen from the metal sites in MOCs have been reported,^{6,9,47,48} which is achieved by introducing additional catalytic metal sites onto the cages. Unfortunately, due to the saturated coordination, the potential catalytic functions of the built-in metal centers of the cage are often hindered or neglected. Therefore, how to integrate and utilize active metal centers on the cage, such as redox-active metal centers,^{49–53} for a catalytic purpose is a challenging step toward biomimetic supramolecular catalysis.

Our group previously reported face-directed metalloligand stepwise assembly of a heterometallic Pd–Fe octahedron cage MOC-42, which exhibits good acid stability with the Fe centers promising for redox-active reactions.²⁶ However, the Pd^{2+} vertices are prone to be reduced in a redox process leading

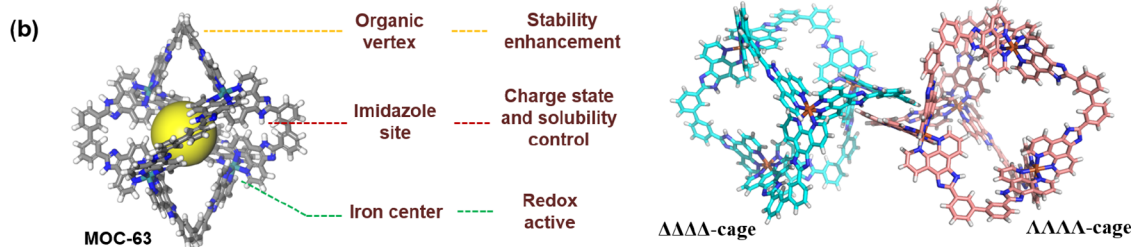
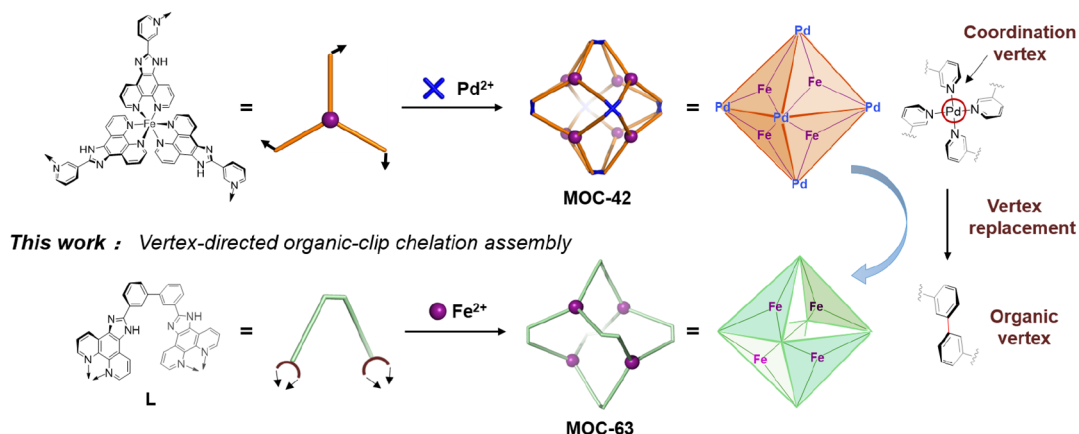
Received: March 11, 2022

Published: May 4, 2022



Scheme 1. (a) Design and Synthetic Route for the $\text{Fe}_4\text{L}_6^{8+}$ Cage (MOC-63) in Comparison with the Pd^{2+} -Coordination Assembled $\text{Pd}_6(\text{FeL}_3)_8^{28+}$ Cage (MOC-42).²⁷ (b) Crystal Structure of MOC-63 Showing Functional Sites and the Stacking Mode of Homochiral $\Delta\Delta\Delta\Delta$ and $\Lambda\Lambda\Lambda\Lambda$ Cages

(a) **Previous work:** Face-directed metalloligand stepwise assembly



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to cage decomposition, and the alkaline conditions popular for Fe-catalyzed reactions⁵⁴ may also result in cage collapse. To avoid these instability drawbacks, herein, we propose a vertex-directed organic-clip chelation assembly method for the construction of a redox-active $\text{Fe}_4\text{L}_6^{8+}$ octahedral cage (MOC-63). As depicted in Scheme 1, this cage is assembled from six bis-bidentate ligands via chelation of four Fe^{2+} ions, which can be regarded as an organic counterpart of MOC-42 through replacing the six 4-connecting Pd^{2+} centers on the vertices by six 2-connecting ditopic L ligands as organic clips. This cage is also analogous to the classic Fujita's $\text{Pd}_6\text{L}_4^{12+}$ cage as a porous and hollow cage with four octahedron faces open for facile guest exchange,⁴² but the metal and organic parts are reversed between vertices and faces. As a consequence, MOC-63 is sufficiently stable in both acidic and basic solutions and avoids cage decomposition from Pd^{2+} reduction. The imidazole moieties on the cage serve as dual proton donors and acceptors to make MOC-63 able to be both protonated and deprotonated simply by varying pH values of solution, thus tuning the solvation and total charge state of MOC-63 to adjust its solubility in different solvents. In addition, the cage stabilization derived from the mechanical coupling between connected vertices and faces²⁶ endows MOC-63 with appropriate dynamic behavior to allow partial dissociation of $\text{Fe}-\text{N}\text{ON}$ chelation bonds while retaining the cage shape, thus rendering this cage robust redox activity toward radical-based dehydrogenation catalytic reactions.

RESULTS AND DISCUSSION

The Assembly and Crystal Structure of MOC-63. The clip-like bis-bidentate ligand L (3,3'-bis(1H-imidazo[4,5-f]-[1,10]-phenanthroline-2-yl)-1,1'-biphenyl) was prepared in a

high yield from the Debus–Radziszewski imidazole synthesis reaction from [1,1'-biphenyl]-3,3'-dicarbaldehyde, 1,10-phenanthroline-5,6-dione, and ammonium acetate, which chelates Fe^{2+} ions in a 6:4 ratio leading to quantitative formation of the $[\text{Fe}_4\text{L}_6](\text{BF}_4)_8$ cage designated as MOC-63. For comparison, the componential Fe complexes $\text{Fe}(\text{PIP})_3(\text{BF}_4)_2$ and $\text{Fe}(\text{Phen})_3(\text{BF}_4)_2$ were also prepared (Scheme 1 and Figures S1–S10).

The single-crystal structure analysis has been performed based on the X-ray diffraction data with modest quality (Tables S1 and S2). Due to the presence of large voids and badly disordered solvents/anions, the reflections in the high θ angle are too weak to obtain full completeness and good data/parameter ratios. However, the satisfactory refinement results are sufficient for the cage structure determination, which discloses an octahedral $\text{Fe}_4\text{L}_6^{8+}$ cage comparable with Fujita's $\text{Pd}_6\text{L}_4^{12+}$ cage with six organic clips acting as vertices and four tris-chelating $\text{Fe}(\text{N}\text{ON})_3$ moieties alternatively occupying four of the eight octahedron faces, leaving the rest of the four faces open for guest exchange. The cage size can be estimated by a vertex-to-vertex separation of 3.0 nm or a face-to-face separation of 2.3 nm. Due to a chirality transfer effect through the organic clip, four Fe centers in each cage exhibit homochiral configurations ($\Delta\Delta\Delta\Delta$ or $\Lambda\Lambda\Lambda\Lambda$), while two $\Delta\Delta\Delta\Delta$ and $\Lambda\Lambda\Lambda\Lambda$ cage enantiomers cocrystallize in a single crystal via $\pi-\pi$ stacking to form a racemic solid (Scheme 1b).

Solution Dynamics and Charge Tunability. Formation of the $\text{Fe}_4\text{L}_6^{8+}$ cage structure in solution is evident from measurements of electrospray ionization–high-resolution mass spectrometry (ESI–HR-MS, Figure 1a and Figure S18), which reveals the formula of a series of $[(\text{Fe}_4\text{L}_6 \cdot x\text{H}^+)^{(8-x)+} + y(\text{BF}_4^-)]^{(8-x-y)+}$ species owing to the deprotonation of the

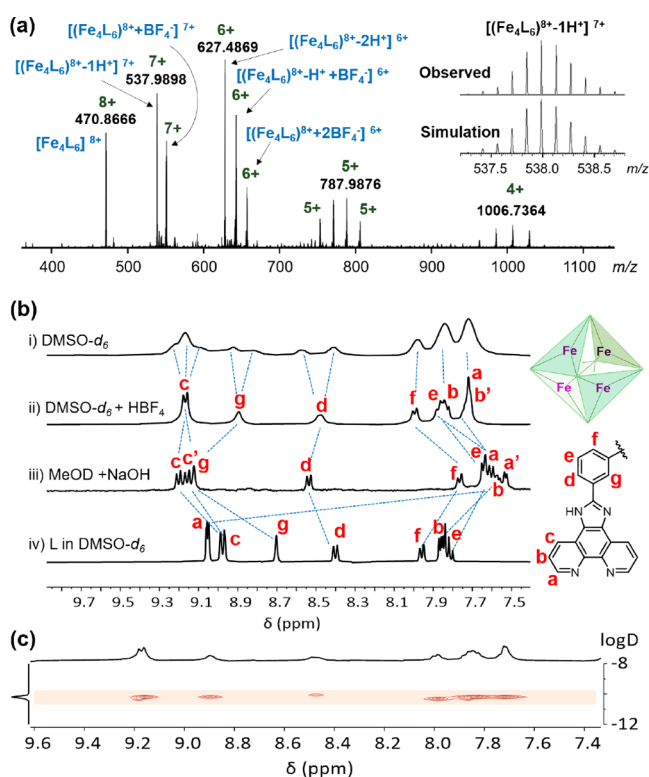


Figure 1. (a) ESI–HR–MS spectra of MOC-63 under neutral conditions in DMSO/CH₃CN (1/20, v/v). (b) Partial ¹H NMR spectra (400 MHz, 298 K) of (i) MOC-63 in DMSO-*d*₆, (ii) MOC-63 in DMSO-*d*₆ added with 2.5 equiv of HBF₄, (iii) MOC-63 in CD₃OD added with 50 equiv of NaOH, and (iv) L in DMSO-*d*₆. (c) ¹H DOSY spectra of MOC-63 in DMSO-*d*₆ containing 2.5 equiv of HBF₄.

exchangeable protons on imidazole motifs, identified by the experimental isotopic patterns well-matched with simulated results. The ¹H NMR signals of MOC-63 are broadened with proton resolutions indistinguishable in DMSO-*d*₆ (Figure 1b), which is common for the large cage structures partly due to their slow tumbling motions on the NMR timescale as observed from reported examples.^{13,14,55} Nevertheless, the appearance of only one set of proton signals indicates that there is a sole thermodynamically stable structure in solution, which is confirmed by ¹H DOSY measurement where one diffusive solution species was observed with the molecular size estimated at 2.6 nm in diameter, in agreement with the single-crystal structural analysis (Figure S16).

To further explore the reason of cage proton signal broadness that leads to ambiguous ¹H NMR resonances that hinder a detailed understanding of the solution structure, we tried the VT-NMR method at elevated temperatures. However, the broad signals show little improvement along with a temperature increase in DMSO-*d*₆ (Figure S11), while the addition of D₂O causes sharpening of signals with better resolved resonances at elevated temperatures (Figure S12).²⁷ These findings suggest that the signal broadness may also relate with cage solvation conditions that presumably influence the dynamic motions of the cages due to potential cage–solvent and cage–cage intermolecular interactions. Considering that the charge state series of MOC-63 observed in ESI–HR–MS are attributable to different deprotonation species, we suspect that the multiple deprotonation states might be another significant factor that interferes with the resonances

of ¹H NMR. To test this conjecture, ¹H NMR measurements are carried out under acidic and alkaline conditions to supply additional H⁺ or to grab the exchangeable H⁺ on the cage, in order to simplify the cage constituents in solution. As expected, in both conditions, MOC-63 presents a set of well-discriminated resonances (Figure 2b), which are able to be

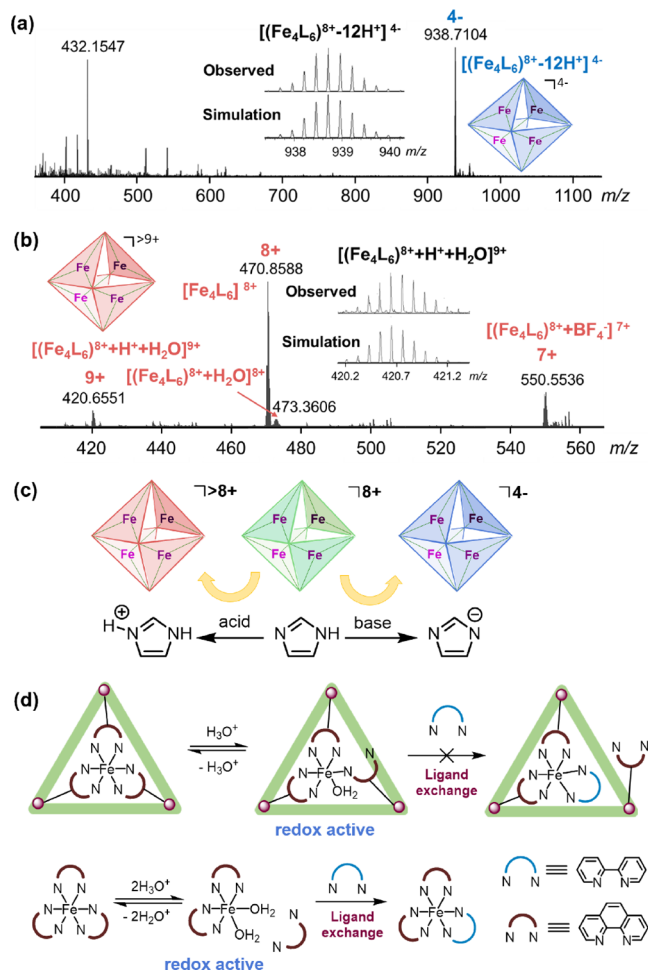


Figure 2. (a) ESI–HR–MS spectra of MOC-63 (0.05 mM) under alkaline conditions in CH₃OH (0.05 M NaOH). (b) ESI–HR–MS spectra of MOC-63 (0.05 mM) under acidic conditions in CH₃CN (0.03 M HBF₄). (c) Illustration of acid–base-modulated charge state conversions via imidazole protonation and deprotonation. (d) Illustration of the dynamic coordination of Fe centers in MOC-63 (upper) and Fe(Phen)₃ (lower) showing distinct dissociation and ligand exchange behaviors to expose redox-active Fe centers.

accurately assigned through ¹H–¹H COSY and ¹H–¹³C HSQC spectra (Figures S13–S15). According to previous reports,^{7,56} the highly positive charges on the cage are able to induce a stepwise loss of the exchangeable protons at the imidazole sites. Therefore, the variable charge states and protonation states will result in not only an increase in the chemical constitution complexity of the cages but also largely affected intermolecular interactions such as hydrogen bonding between solvents and imidazoles, thereby complicating the ¹H NMR resonances. In acidic conditions, formation of differently deprotonated species is suppressed by the excess H⁺ hydrons, which further protonate imidazoles to prevent proton tautomerization. The calculated cage diameter from DOSY under acidic conditions is 2.9 nm, which is slightly larger than

the result in neutral conditions (Figure 1c). In contrast, in a basic solution, the cages may lose all imidazole protons to give 4− charged species, thus making the ^1H NMR resonances much better distinguishable without interference from exchangeable protons. The size of such a fully deprotonated cage is 2.4 nm from DOSY (Figure S17), which is generally smaller than the protonated cage as anticipated.

The charge tunability of MOC-63 under acidic and alkaline conditions has been tested by ESI–HR-MS (Figure 2a,b). In an alkaline solution, the 4− charged isotopic pattern is dominant in the MS spectrum, which corresponds to an $\text{Fe}_4\text{L}_6^{4-}$ cage losing all 12 imidazole-NH protons through complete deprotonation ($m/z = 938.7104$, Figure 2a,c and Figure S19). On the contrary, in an acidic solution, only high charge states of MOC-63 are obtained for a 9+ charged species at $m/z = 420.6551$, two 8+ species at $m/z = 470.8588$ and 473.3606 , and a 7+ species at $m/z = 550.5536$, corresponding to the $[(\text{Fe}_4\text{L}_6)^{8+} + \text{H}^+ + \text{H}_2\text{O}]^{9+}$, $[(\text{Fe}_4\text{L}_6)^{8+} + \text{H}_2\text{O}]^{8+}$, and $[(\text{Fe}_4\text{L}_6)^{8+} + \text{BF}_4^-]^{7+}$ cage series, respectively, showing no deprotonated species as observed in neutral conditions (Figure 2b,c and Figure S20). These results imply that MOC-63 may undergo apparent charge variation through H^+ transfer via 12 imidazole proton donors and acceptors, thus offering a wide range of charge tunability from 4− to higher than 8+ (Figure 2c) to give a rare example of a charge tunable cage,⁵⁷ in agreement with the NMR study in the acidic and alkaline conditions.

It is worth noting that the H_2O -binding species can be observed in the mass spectrum in acidic conditions (Figure 2b), which retains the cage integrity without the loss of any components. We speculate that these H_2O -binding peaks may relate with the partial ligand dissociation via $\text{Fe-N}\text{N}$ chelation rupture to liberate one coordination site on the Fe center for rapid water exchange (Figure 2d), following a well-established dissociation and racemization mechanism for the tris-chelating $\text{Fe}(\text{N}\text{N})_3$ complexes that is facilitated by acidity and solvent replacement.^{58–61} To prove such coordination dynamics, we further investigate the cage stability in solution. It is well-known that, due to the lability of the Fe-N bond, most $\text{Fe}(\text{N}\text{N})_3$ chelating complexes are unstable in solution, undergoing fast ligand dissociation and racemization processes through one or more Fe-N bonds breaking.^{58–60} However, benefiting from a cage stabilization effect²⁶ that greatly slows down the dissociation rate and inhibits racemization through mechanical coupling of multiple metal centers^{62,63} bridged by ligands to enforce cage configurational rigidity, MOC-63 is robust enough to retain cage integrity under the treatment of an acid and a base, keeping the NMR proton profiles nearly unchanged for more than one week (Figures S21 and S22). For comparison, the related fragmental $\text{Fe}(\text{PIP})_3^{2+}$ and $\text{Fe}(\text{Phen})_3^{2+}$ complexes show ligand dissociation immediately when an acid is added (Figure S23). Moreover, we carry out competitive ligand replacement experiments to check the coordination dynamics by applying 2,2′-bipyridine as a competitor to exchange the ligands in MOC-63 and $\text{Fe}(\text{Phen})_3^{2+}$ (Figure 2d and Figure S24). As seen from NMR monitoring, MOC-63 exhibits strong resistance to this competitive exchange because neither the free L ligand nor the Fe-bpy compound is found in solution. In contrast, the addition of the bpy competitor obviously results in ligand dissociation from $\text{Fe}(\text{Phen})_3$ to form new Fe-bpy-related complexes. The above results manifest the good robustness of MOC-63 to keep the overall cage connectivity when one of the

branching $\text{Fe}(\text{N}\text{N})_3$ chelation complexes undergoes solvent or substrate exchange. This implies that partial dissociation of $\text{Fe-N}\text{N}$ chelation may occur via rupture of a single Fe-N bond⁶⁰ to allow rapid solvent exchange on the opened coordination site,⁵⁸ making it feasible to observe water-binding cage species in the mass spectrum (Figure 2b). Such coordination dynamics is known to be able to accelerate in acidic conditions to expose redox-active Fe centers (Figure 2d) that is necessary for a catalytic purpose (*vide infra*).^{54,60,61,64,65} To our delight, the charge state changes also provide a solvation effect to adjust the solubility of MOC-63 in different solvents (Table S3), which will facilitate a simple catalyst recycling process by switching the solution acidity through pH changes.

Dehydrogenation Catalysis and Recycling Performance. Fe complexes are known for their redox activity to initiate radical reactions,^{54,64,66,67} and dehydrogenation of N-heterocycles is of fundamental importance in organic transformations.^{68,69} Therefore, the dehydrogenation of 1,2,3,4-tetrahydroquinoline (THQL) was chosen as a model reaction to test the potential of MOC-63 as a redox-active catalyst for this dehydrogenation reaction (see the Supporting Information). The solvent screening experiments indicate that water is essential in this catalytic reaction with an appropriate proportion (Figures S25–S28). In an optimized $\text{DMSO-d}_6\text{:D}_2\text{O}$ (3:1) (v/v) mixture, the THQL dehydrogenation reaction achieves a high yield up to 98% in 7 h at room temperature (Figure S29). The control experiments performed using two fragmental Fe complexes $\text{Fe}(\text{Phen})_3^{2+}$ and $\text{Fe}(\text{PIP})_3^{2+}$ with equivalent Fe amounts unveil much lower conversions in comparison with that of MOC-63 under the same conditions (Figure 3a and Figures S30 and S31). The similar rate acceleration effect by cage catalysis is also confirmed by dehydrogenation reactions of 7-phenyl- and 6-bromo-substituted THQL derivatives (Figures S32–S37). Moreover, the reaction in the absence of TBHP (*tert*-butyl hydroperoxide) shows no detectable product (Figure S38).

A striking feature in the kinetic profile of MOC-63 catalysis is the observation of an induction period followed by an obvious acceleration step, in contrast to the almost linear reaction rate of $\text{Fe}(\text{Phen})_3^{2+}$ and $\text{Fe}(\text{PIP})_3^{2+}$ (Figure 3a). This means that the generation of reactive species in the case of MOC-63 is slower than that of simple Fe complexes at the early stage, suggesting that it is relatively hard to activate the catalytic sites on the cage to produce intermediates, probably due to the cage stabilization effect and only partial dissociation of the ligand on Fe centers mentioned above. However, as long as the conversion of the intermediates starts, the overall reaction proceeds much faster (Figure 3c). Surprisingly, the reaction progresses are significantly altered in the presence of an acid or a base (Figure 3b and Figures S39–S41). In an acidic condition, the reaction is greatly sped up with a much shorter induction period. When 0.5 mM HOTf is added into the reaction system, the substrates are completely converted within 2 h with an induction period of half an hour, which can be even shortened to 1.5 h for a full conversion and 20 min for induction by adding 1 mM HOTf. On the contrary, the presence of NaOH nearly suppresses the catalytic performance of MOC-63. Therefore, an acid–base regulatory halting-and-restarting reaction process can be established by controlling the solution acidity (Figures S42 and S43). When 0.5 mM NaOH is added, the reaction is retarded; meanwhile, the

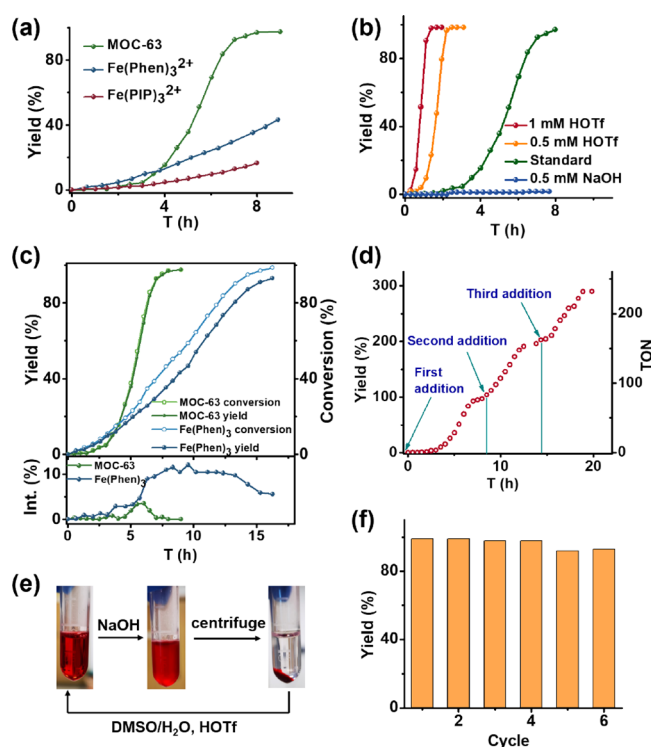


Figure 3. (a) Reaction progresses showing yields of quinolone as a function of time catalyzed by MOC-63, fragmental Fe complexes $[\text{Fe}(\text{PIP})_3](\text{BF}_4)_2$ and $[\text{Fe}(\text{Phen})_3](\text{BF}_4)_2$ in $\text{DMSO}-d_6/\text{D}_2\text{O}$ ($v/v = 3/1$) solution at 298 K. (b) Reaction progresses showing yields of quinolone as a function of time catalyzed by MOC-63 in different acid–base conditions. (c) Comparison of the yield of quinolone and conversion of THQL catalyzed by MOC-63 and $[\text{Fe}(\text{Phen})_3](\text{BF}_4)_2$, showing the fractions of intermediates plotted below (fraction of intermediate = conversion – yield). (d) Reaction progresses of three continuous runs by addition of THQL and TBHP three times into the same reaction solution without isolation of the product and the catalyst. (e) Photograph of the cage recycling process through acid–base switching. (f) Recycling performance of MOC-63 on repeated dehydrogenation reactions six times.

sequential addition of 0.5 mM HOTf can reinitiate the reaction progress.

Approximate zero-order fittings of reaction kinetics reveal reaction rates (k) of 1.49 and 0.44 $\text{mmol}\cdot\text{L}^{-1}\cdot\text{h}^{-1}$ catalyzed by $\text{Fe}(\text{Phen})_3^{2+}$ and $\text{Fe}(\text{PIP})_3^{2+}$, respectively (Figures S44 and S45). For MOC-63 under neutral conditions, the induction rate is estimated to be $k_{\text{ind}} = 0.13 \text{ mmol}\cdot\text{L}^{-1}\cdot\text{h}^{-1}$ while the accelerated rate is fitted to be $k_{\text{acc}} = 6.22 \text{ mmol}\cdot\text{L}^{-1}\cdot\text{h}^{-1}$ (Figures S46 and S47). These results indicate the rate enhancement of $k_{\text{acc}}/k_{\text{ind}} = 47.9$ for cage-confined catalysis and the cage acceleration effects of $k_{\text{MOC-63}}/k_{\text{Fe(Phen)3}} = 4.2$ and $k_{\text{MOC-63}}/k_{\text{Fe(PIP)3}} = 14.1$ as compared with $\text{Fe}(\text{Phen})_3^{2+}$ and $\text{Fe}(\text{PIP})_3^{2+}$, respectively. In the presence of 0.5 or 1 mM HOTf, the reaction rates are increased to $k_{\text{acc}} = 19.07$ or 21.57 $\text{mmol}\cdot\text{L}^{-1}\cdot\text{h}^{-1}$, respectively (Figures S48 and S49), which offers even larger acceleration effects, e.g., $k_{\text{MOC-63}}/k_{\text{Fe(PIP)3}} = 43.3$ (0.5 mM HOTf) or $k_{\text{MOC-63}}/k_{\text{Fe(PIP)3}} = 49$ (1 mM HOTf). Therefore, the dehydrogenation catalytic efficiency of MOC-63 is evidently acid–base-dependent, and the acidic media can tremendously expedite the dehydrogenation reaction of THQL. Owing to the fast guest exchange behavior via open windows of MOC-63 (Figure S66), no reaction inhibition is observed in the catalytic reaction progress. The reaction can be

well-continued through successive addition of the oxidants and substrates after every complete conversion, giving turnover numbers of 233 after three runs of reactions in 20 h (Figure 3d).

Interestingly, as the recovery of the catalyst is troublesome for homogeneous catalysis, the solubility tuning of MOC-63 in different solvents (Table S3) as a result of acid–base regulatory charge variation provides an easy handling method for recycling of the MOC-63 catalyst (Figure 3e). When adding 2 equiv of NaOH solution into the reaction mixture, the pH change of the solvent results in MOC-63 precipitation immediately to give a suspension, which can be centrifuged to recover the catalyst. After washing the isolated MOC-63 with water, it can be redissolved in $\text{DMSO}-\text{H}_2\text{O}$ for the second reaction run. By applying this recycling process, we test the lasting catalytic performance of MOC-63 on THQL dehydrogenation for six cycles (Figure 3f), which indicates that the reactivity of MOC-63 is well-retained for recycling catalysis.

We further test the substrate scope of this catalytic reaction under the standard reaction condition (Table 1 and Figures

Table 1. Substrate Scope of MOC-63-Catalyzed Dehydrogenation Reactions of THQL Derivatives under Standard Reaction Conditions for 12 h

R 	
	97%
	85%
	96%
	63%
	62% ^a
	97%
	96%
	86%
	86% ^a
	95%
	35% ^a
	96%
	63%
	88%

^aYield after 24 h.

S50–64). Substrates containing methyl, phenyl, and halide groups exhibit generally good to excellent reactivity with yields in the range of 86–97%. The conversions of substrates having the same substituting group are varied with dependence on their substituent positions, while different substituting groups at the same position show diversified conversions. Relatively low reactivity is observed for substrates containing O-involved substituents, e.g., ester and methoxyl groups, no matter whether they have electron-withdrawing or electron-donating character. A moderate yield is found for indoline with a five-membered ring, while excellent yield is obtained for 7-phenyl-THQL with a large conjugation. These catalytic results indicate a wide scope of substrates for this cage-defined supramolecular catalysis; however, the catalytic behavior in relation to different substituents is distinct from other reported dehydrogenation catalysis of N-heterocycles.^{70,71} A plausible reason is that the host–substrate interactions may affect the catalytic efficiency. For example, the substituents containing an O atom may be more influenced due to its strong H-bonding

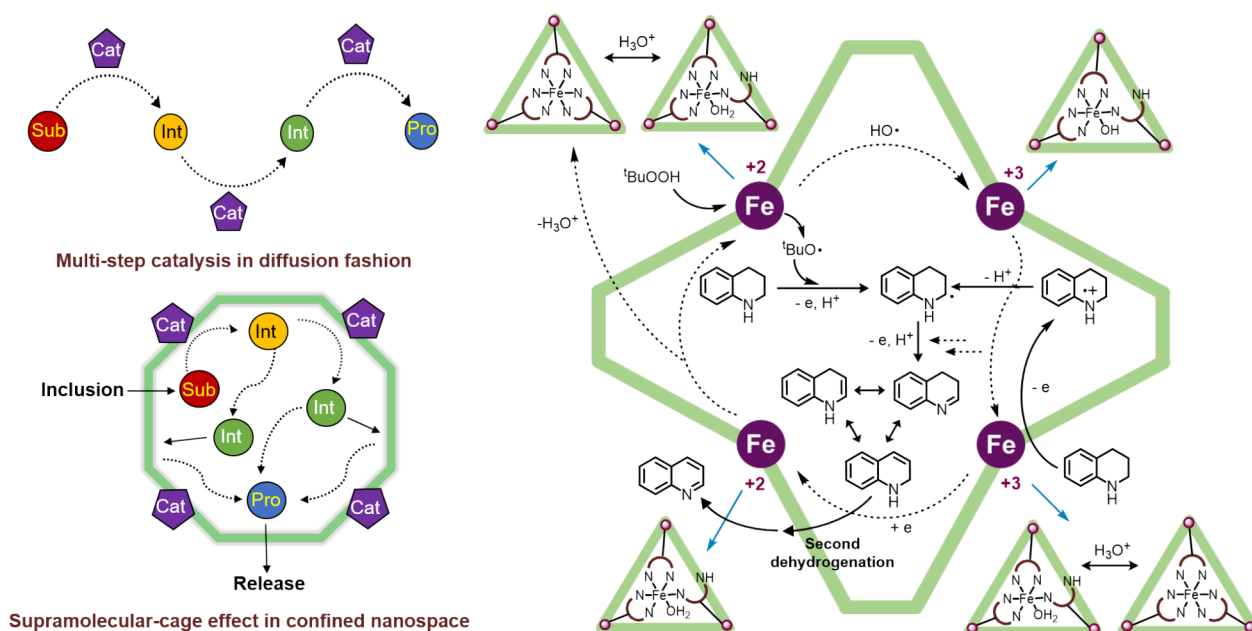


Figure 4. Illustration of cage-confined catalysis of the multistep radical reactions with a supramolecular cage effect vs conventional catalysis of the multistep radical reactions with a diffusive process (Sub: substrate; Int: radical intermediate; Pro: product) and the possible reaction mechanism involving continuous transformation of THQL into radical intermediates and tautomers in a cage-confined nanospace via synergism of multiple redox-active Fe centers.

ability to interact with imidazole-NH sites on the outside of the cage, while smaller-sized indoline may be less fit with the cage than the larger conjugating 7-phenyl-THQL. Such a slight host–guest interaction difference presumably has an impact on the guest inclusion dynamics, thus affecting the acceleration effect on different substrates with varied substitution positions and groups as mentioned above.

Mechanism Insight. As a fundamental property of cages, the host–guest chemistry of MOC-63 is studied first. One piece of evidence for the guest encapsulation is obtained by HR-MS measurement for the 7-Ph-THQL/MOC-63 clathration, which shows peaks of $[(\text{Fe}_4\text{L}_6)^{8+} \cdot 3\text{H}^+ + (7\text{-Ph-THQL})]^{5+}$ and $[(\text{Fe}_4\text{L}_6)^{8+} \cdot 3\text{H}^+ + 2(7\text{-Ph-THQL})]^{5+}$, corresponding to the 1:1 and 1:2 binding species, respectively (Figure S65). This result suggests that at least two 7-Ph-THQL guests could be hosted in the cage cavity. The binding constants for the host–guest interactions are estimated for representative guests of THQL and its 2-methyl-, 7-phenyl-, and 6-bromo-substituted derivatives by means of NMR titrations (Figures S66–S73). The open cavity renders a shielding effect to the fast exchanged guests, thereby shifting the proton resonances of guests progressively upfield. In all cases, the best curve fitting with the titration data is obtained using a 1:2 binding model of Bindfit (supramolecular.org),⁷² giving binding constants of THQL to be $K_{11} = 115.00 \pm 0.18$ and $K_{12} = 28.74 \pm 0.05 \text{ M}^{-1}$, those of 2-methyl-THQL to be $K_{11} = 72.54 \pm 0.11$ and $K_{12} = 18.14 \pm 0.03 \text{ M}^{-1}$, those of 6-bromo-THQL to be $K_{11} = 202.20 \pm 0.19$ and $K_{12} = 50.50 \pm 0.05 \text{ M}^{-1}$, and those of 7-phenyl-THQL to be $K_{11} = 374.60 \pm 0.59$ and $K_{12} = 93.70 \pm 0.15 \text{ M}^{-1}$. These results imply that guest inclusion into MOC-63 may approximate a noncooperative behavior. The fast kinetics of host–guest interactions might be subjected to the solvent influence owing to the well-known hydrophobic effect.^{73–75} Increasing the DMSO:H₂O ratio may facilitate solubility and diffusion of organic guests, thus promoting the reaction rate but counteracting the hydrophobic effect. On the contrary,

decreasing the DMSO:H₂O ratio may enhance guest binding while retarding fast guest exchange leading to product inhibition.⁷⁵ This may be a reason why the optimized DMSO:D₂O = 3: 1 (v/v) ratio is competent for the efficient catalytic reaction.

Competitive experiments have been performed to elucidate the role of the cage cavity in this reaction in the presence of sodium pyrene-1,3,6,8-tetrasulfonate (PTSNa) as a strong competing guest (Figures S74–S79), which shows a much larger binding constant ($18,121 \pm 369 \text{ M}^{-1}$) and PTS^{4-} /MOC-63 binding species on HR-MS. Under the standard condition, the reaction rate catalyzed by the cage is significantly decreased after the addition of 1 equiv of the competing guest PTSNa. For comparison, the added competitor PTSNa does not reduce the reaction rate catalyzed by mononuclear $\text{Fe}(\text{Phen})_3$, while slightly promotes the reaction instead. These results suggest that the acceleration effect mainly comes from the guest encapsulation by the cage cavity; however, contributions from the catalysis on the outside surface and cage windows are always possible and should not be ruled out.^{76–80}

To determine the reaction nature of dehydrogenation, a radical trapping experiment is performed by using TEMPO (2,2,6,6-tetramethylpiperidinyloxy) as radical-scavenging agent, which confirms a radical process. When TEMPO is added, the dehydrogenation reaction is nearly suppressed, while a TEMPO adduct as a radical intermediate is detected by MS (Figures S80 and S81). Therefore, the overall THQL dehydrogenation is expected to be achieved through the loss of two electrons and two H^+ via a multistep radical process involving a series of radical and tautomeric intermediates (Figure 4), where the substrates have to travel and collide with different catalytic centers and intermediates to complete a succession of reactions.

The initiation step may require a single electron transfer (SET) from the substrate to produce a radical intermediate,

which should involve $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox conversion mediated by 'BuOOH (TBHP). Cyclic voltammetry (CV) measurements of MOC-63 elucidate a quasi-reversible wave at $E_{1/2} = 1.084$ V (vs SCE, Figure S82) attributable to the $\text{Fe}^{3+}/\text{Fe}^{2+}$ potential, which is comparable with the potentials of reported Fe–polypyridyl complexes⁶⁷ and confirms the possibility to generate the amine radical cations ($E_{1/2} = 0.74$ V vs SCE).⁸¹ Furthermore, when 30 mM HOTf is added, the $\text{Fe}^{3+}/\text{Fe}^{2+}$ potential is slightly reduced to $E_{1/2} = 1.048$ V with the ΔE_p values greatly decreased from 188 to 109 mV, which indicates a much improved reversibility. Therefore, the presence of an acid is in favor of $\text{Fe}^{3+}/\text{Fe}^{2+}$ charge interconversion. As TBHP could not directly oxidize $\text{Fe}(\text{N}\text{N})_3$ moieties due to potential mismatch, it is likely that partial dissociation of the ligand⁶⁰ occurs under acidic conditions to form a H_2O -binding species of MOC-63 as the redox-active state⁶⁴ (Figure 2d), which can easily exchange with the radical species to initiate the reaction. This hypothesis also accounts for the reason why reactions are sped up at high concentrations of an acid while suppressed in the presence of a base.

Based on the above experimental results and the reported works of $\text{Fe}(\text{N}\text{N})_3$ -type catalysts^{61,64} in which the $\text{Fe}^{3+}/\text{Fe}^{2+}$ charge state alternation and ligand dissociation play a pivotal role in oxidizing substrates into radical cations, a plausible reaction route is tentatively proposed as illustrated in Figure 4 and Figure S83. A key step may be the rupture of a single Fe–N N chelation bond⁶⁰ to generate H_2O -binding cage species with better redox activity. Under the action of hydronium ions, one ligand on an Fe center can reversibly and partially dissociate and recombine rapidly, thus activating the redox Fe center. Subsequently, this water-binding $[\text{Fe}(\text{II})\text{N}_5\text{-H}_2\text{O}]$ site is exchanged and oxidized by TBHP to form an $[\text{Fe}(\text{III})\text{N}_5\text{-OH}]$ center and a 'BuO· radical. The generated 'BuO· reactive species is capable of inducing THQL to give an alkyl radical,^{82,83} while the $[\text{Fe}(\text{III})\text{N}_5\text{-OH}]$ center is able to seize a H^+ to turn to $[\text{Fe}(\text{III})\text{N}_5\text{-OH}_2]$ and undergo ligand recombination to establish equilibrium with $[\text{Fe}(\text{III})(\text{N}\text{N})_3]$. In turn, the strongly oxidative $[\text{Fe}(\text{III})(\text{N}\text{N})_3]$ may promote the production of the amine radical cations⁶⁴ from THQL, which will be converted to the alkyl radicals with the aid of $[\text{Fe}(\text{III})\text{N}_5\text{-OH}]$ centers to further undergo the second dehydrogenation process via SET and H^+ abstraction processes again. Therefore, the multistep radical reactions proceeding in the cage-confined nanospace of MOC-63 have an induction stage to generate a series of radical intermediates and catalytic Fe centers, which is distinct from the reactions catalyzed by mononuclear $\text{Fe}(\text{N}\text{N})_3$ complexes without the cage stabilization effect. However, once the Fe centers on the cage are activated and the radical intermediates are initiated inside the cage, the multistep conversions of the substrates into the first and successively second dehydrogenation products will be accelerated. This possible reaction progress is also supported by the observation that the continued runs of reactions in the same media without isolation of the cage catalyst do not show an obvious induction period anymore (Figure 3d).

We speculate that the acceleration of multistep radical reactions catalyzed by MOC-63, in comparison with the control experiments of normal $\text{Fe}(\text{N}\text{N})_3$ complexes, is due to a supramolecular cage effect arisen from the cage-confined nanospace integrating four Fe centers for synergism of multiple catalytic sites and radical conversions (Figure 4). Different from the conventional homogeneous catalysis of $\text{Fe}(\text{N}\text{N})_3$ complexes in free diffusion and collision fashion, the cage-

confined catalysis of the multistep radical reactions proceeds in a porous cage solution where the cages can be regarded as reactive nanoreactors with much more effectiveness because multiple active sites are arranged proximately on the cage wall that can reflect radicals back into more collisions, increasing the mean lifetime and concentration of radical intermediates for efficient conversion before escaping into bulk solution. This situation is strikingly distinguishable from the traditional "solvent cage effect"^{84,85} that usually causes a lower efficiency of free radical reactions in solution than in the gas phase due to trapping of radicals inside the solvent cage that decreases collision probability of free radicals diffused out of the cage. Moreover, as the guest exchange happens typically in an order of the NMR timescale of milliseconds, the robust supramolecular cage is expected to impose much more impact on radical movements than the solvent cage just in a very short-lived transient form ($\sim 10^{-11}$ s), leading to a high conversion ratio of intermediates once the substrates are transformed into intermediates via initiation and concentration of radicals by densely surrounded redox-active Fe centers on the cage wall.

To estimate the conversion rates of intermediates in the cage-confined catalysis versus the control experiments, we calculate the survival intermediates through subtracting the instantaneous yield from conversion (Figure 3c). By comparing the fraction of intermediates detectable in the progress of reactions, we can see that the intermediates are consumed faster in the presence of the cage, in agreement with the observation that the radical reaction is expedited in the cage-confined catalysis. More direct pieces of evidence are obtained by NMR and EPR measurements of the non-paramagnetic intermediates and radical intermediates individually at a certain stage when the same yield is achieved. The NMR signals of the observed intermediates in the reactions catalyzed by MOC-63 are significantly lower than those of intermediates in reactions catalyzed by $\text{Fe}(\text{Phen})_3^{2+}$ (Figure S84). Moreover, when adding the trapping reagent DMPO (5,5-dimethyl-1-pyrroline *N*-oxide) into the reaction solutions at the same yield, the EPR signals of the trapped radicals in the cage reaction are significantly weaker than those in the $\text{Fe}(\text{Phen})_3^{2+}$ reaction (Figure S85). These results definitely denote a higher total concentration of radical intermediates that survived in the reaction progress when using $\text{Fe}(\text{Phen})_3^{2+}$ as a catalyst in comparison with the cage catalyst, suggesting that the cage-confined catalysis proceeds in a more efficient and faster way to consume the intermediates in a relatively short period of time.

CONCLUSIONS

A multifunctional Fe-based imidazole-containing $\text{Fe}_4\text{L}_6^{8+}$ supramolecular cage is developed by adopting organic clips as vertices instead of regular coordination metal vertices, which greatly enforces robust solution dynamics and stability under reaction conditions involving acids, bases, and a redox process. A wide range of charge tunability is achieved through multiple H^+ transfer to adjust the cage solubility in varied solvents and improve ^1H NMR resonances for a better understanding of cage structural information in solution. Acid–base-dependent redox activities are established for supramolecular catalysis of the dehydrogenation reaction, in which the porous cage solution endows cage nanoreactors with a supramolecular cage effect to accelerate multistep radical reactions inside cage nanospaces, in striking contrast to the normal solvent cage effect that otherwise retards the free radical reaction in

solution. The synergistic actions are observed from multi-Fe centers that are assembled in the cage-confined nanospace with a high local density of redox-active catalytic centers and a concentration of reactive intermediates and tautomers for efficient conversion in a relatively short period of time. Moreover, excellent continuing and recycling performance of the cage catalyst is achievable, and a convenient catalyst recycling process can be obtained simply through acid–base switching of the reaction solution. This work demonstrates a methodology for utilizing the dynamic nature of the saturated coordination sites on the cage structure to create reaction activity, which may shed light on how to push supramolecular cage chemistry toward more functionalities and applications.

■ EXPERIMENTAL DETAILS

Synthesis of L. Into a round-bottom flask containing 30 mL of acetic acid were added [1,1'-biphenyl]-3,3'-dicarbaldehyde (210 mg, 1 mmol), 1,10-phenanthroline-5,6-dione (440 mg, 2.1 mmol), and ammonium acetate (2 g, 26 mmol). The reaction mixture was heated to reflux for 6 h and then cooled to room temperature to pour the yellow precipitate into 100 mL of deionized water. Ammonium hydroxide was added to the mixture to adjust the pH value to 6. The solid product was filtered off, washed with water (2 × 50 mL) and ethanol (50 mL) and dried under vacuum to give a pale orange solid (501 mg, 85%). ¹H NMR (400 MHz, DMSO-*d*₆): δ 9.05 (dd, *J* = 4.2, 1.6 Hz, 4H), 9.01–8.94 (m, 4H), 8.70 (s, 2H), 8.40 (d, *J* = 7.8 Hz, 2H), 7.96 (d, *J* = 7.7 Hz, 2H), 7.90–7.78 (m, 6H). ¹³C NMR (151 MHz, DMSO-*d*₆): δ 150.84, 148.34, 144.12, 141.24, 131.25, 130.30, 130.20, 128.69, 126.26, 125.11, 123.79.

Synthesis of MOC-63. L (400 mg, 0.68 mmol) was added into a 50 mL round-bottom flask containing 20 mL of DMSO and heated to 80 °C until all L was dissolved; then, 1 mL of Fe(BF₄)₂·6H₂O (152.8 mg, 0.45 mmol) in DMSO solution was slowly added to the mixture and heated for 24 h. The resulting red-violet solution was poured into saturated NaBF₄ water solution to give a red precipitate and washed with deionized water and ethanol and dried under vacuum to give a deep red solid (497 mg, 90%). ¹H NMR (400 MHz, DMSO-*d*₆ added with 2.5 equiv of HBF₄): δ 9.17 (d, 2H), 8.90 (s, 2H), 8.48 (s, 2H), 7.99 (d, 2H), 7.90–7.80 (m, 2H), 7.74–7.70 (m, 3H). ¹H NMR (400 MHz, 50 mM NaOH in methanol-*d*₄): δ 9.20 (d, *J* = 8.2 Hz, 1H), 9.16 (d, *J* = 8.4 Hz, 1H), 9.12 (s, 1H), 8.54 (d, *J* = 7.9 Hz, 1H), 7.76 (d, *J* = 7.8 Hz, 1H), 7.60–7.56 (m, 4H), 7.53 (d, *J* = 5.2 Hz, 1H).

Synthesis of [Fe(PIP)₃](BF₄)₂. 2-(Phenyl-3-yl)-1H-imidazo[4,5-*f*][1,10]phenanthroline (PIP, 296 mg, 1 mmol) was dissolved in 10 mL of DMSO to react with Fe(BF₄)₂·6H₂O (112.3 mg, 0.33 mmol), giving a dark red solution. The addition of 100 mL of ethyl acetate resulted in a deep red solid, which was washed with ethyl acetate three times and dried under vacuum (300 mg, 96% yield). ¹H NMR (400 MHz, DMSO-*d*₆): δ 14.40 (br, 1H), 9.12 (d, *J* = 8.1 Hz, 2H), 8.33 (d, *J* = 7.6 Hz, 2H), 7.95–7.77 (m, 2H), 7.70 (dd, *J* = 14.9, 6.6 Hz, 4H), 7.61 (t, *J* = 7.4 Hz, 1H).

Synthesis of [Fe(Phen)₃](BF₄)₂. Phenanthroline (Phen, 210 mg, 1 mmol) was dissolved in 10 mL of CH₃CN to react with Fe(BF₄)₂·6H₂O (112.3 mg, 0.33 mmol), giving a dark red solution. The addition of 50 mL of diethyl ether resulted in a deep red solid, which was washed with diethyl ether three times and dried under vacuum (200 mg, 87% yield). ¹H NMR (400 MHz, DMSO-*d*₆): δ 8.81 (d, *J* = 7.5 Hz, 1H), 8.41 (s, 1H), 7.74 (m, 2H).

Standard Dehydrogenation Reaction. The catalytic reaction was performed with 1,2,3,4-tetrahydroquinoline (THQL) and monitored by ¹H NMR in typical reaction conditions: a 1.25% loading of the cage or a 5% loading of Fe complexes, 5 equiv of TBHP, and DMSO-*d*₆:D₂O = 3: 1 (v/v) solution. The yield was determined by ¹H NMR using DMF as an internal standard at 298 K.

Control Experiments. [Fe(PIP)₃](BF₄)₂ (0.625 mg, 0.55 μmol, 5% loading) and [Fe(Phen)₃](BF₄)₂ (0.43 mg, 0.55 μmol, 5% loading) were dissolved in DMSO-*d*₆ (375 μL) and D₂O (125 μL), and THQL (1.48 mg, 11.1 μmol) and 5 equiv of TBHP (70 wt % in

H₂O, 7.6 μL, 55.5 μmol) were added sequentially. DMF (0.2 μL) was added as an internal standard. The yield was monitored by ¹H NMR at 298 K.

Acid–Base-Modulated Reaction. MOC-63 (0.625 mg, 0.14 μmol, 1.25% loading) was dissolved in DMSO-*d*₆ (375 μL) and D₂O (125 μL), and THQL (1.48 mg, 11.1 μmol) and 5 equiv of TBHP (70 wt % in H₂O, 7.6 μL, 55.5 μmol) were added sequentially. DMF (0.2 μL) was added as an internal standard. A stock solution of HOTf (2.5 μL, 0.1 M in D₂O) was added for acidic conditions, and a stock solution of NaOH (2.5 μL, 0.1 M in D₂O) was added for basic conditions. The yield was monitored by ¹H NMR at 298 K.

Catalyst Recovery. Benefiting from the acid modulation nature of MOC-63, the catalyst can be easily recovered by adjusting the acidity of the solution. The typical procedure was as follows: under typical reaction conditions [1.25% loading of MOC-63 (0.625 mg, 0.14 μmol), THQL (1.48 mg, 11.1 μmol), TBHP (70 wt % in H₂O, 7.6 μL, 55.5 μmol), HOTf (1 μmol), and DMSO-*d*₆:D₂O = 3: 1], NaOH water solution (25 mM, 100 μL, 2 equiv) was added into the solution after 2 h of the reaction at room temperature, and the precipitate was collected by centrifugation and washed with water twice. The precipitate was then redissolved in the reaction solution to proceed to the next cycle.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.2c02692>.

Materials and methods, spectral characterization, and single-crystal structure crystallography (PDF)

Accession Codes

CCDC 2142981 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the NKRD Program of China (2021YFA1500401), the NSFC projects (21821003, 21890380, 21720102007, 21771197, and 22003079), the LIRTP of Guangdong Pearl River Talents Program (2017BT01C161), and the NSF of Guangdong Province (2021A1515010298).

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